Zara Cromwell

Contact Information |
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Professional Summary

Highly analytical and results-oriented Computer Vision Specialist with 5+ years of experience at Synthetica Global, leveraging cutting-edge AI/ML techniques to develop and deploy innovative solutions. Proven ability to design, implement, and optimize complex computer vision algorithms, utilizing deep learning frameworks such as TensorFlow and Hugging Face Transformers, and cloud platforms like AWS. Expertise in Java, Go, and Python, coupled with a strong foundation in mathematics and computer science from MIT, enables me to tackle challenging problems and deliver impactful results. Seeking a challenging role where I can contribute my expertise to advance the field of Artificial Intelligence.

Education

Massachusetts Institute of Technology (MIT), Cambridge, MA | Master of Science in Electrical Engineering and Computer Science (MSEE) | May 2018 | GPA: 3.9/4.0 | Thesis: "Real-time Object Detection using Optimized Convolutional Neural Networks"

Massachusetts Institute of Technology (MIT), Cambridge, MA | Bachelor of Science in Computer Science (BSc) | May 2016 | GPA: 3.8/4.0 | Minor: Mathematics

Technical Skills

Programming Languages: Java, Go, Python, C++ (proficient), JavaScript (intermediate)

Deep Learning Frameworks: TensorFlow, PyTorch, Keras, Hugging Face Transformers

Cloud Computing: AWS (EC2, S3, Lambda, RDS), Google Cloud Platform (GCP) (Basic), Azure (Basic)

Computer Vision Libraries: OpenCV, scikit-image

Databases: SQL, NoSQL (MongoDB)

Other: Docker, Kubernetes, Git, Agile methodologies

Professional Experience

Computer Vision Specialist | Synthetica Global | Boston, MA | June 2018 – Present

- Led the development and deployment of a real-time object detection system for autonomous vehicles, resulting in a 15% improvement in accuracy and a 10% reduction in processing time.
- Designed and implemented a novel deep learning model for image segmentation, achieving state-of-the-art results on a challenging medical image dataset.
- Developed and maintained robust and scalable machine learning pipelines using AWS services, ensuring efficient data processing and model deployment.
- Mentored junior engineers in best practices for software development and machine learning.
- Contributed to the development of a new product feature using Go for improved performance and scalability.
- Collaborated with cross-functional teams (product, engineering, design) to deliver high-quality products.

AI/ML Projects

Real-time Object Detection for Autonomous Vehicles (Synthetica Global)

- Developed a real-time object detection system using YOLOv5 and TensorFlow, achieving 95% accuracy on a diverse dataset of road scenes.
- Optimized the model for deployment on edge devices using TensorFlow Lite, reducing inference time by 50%.
- Deployed the system to a fleet of autonomous vehicles, resulting in a significant improvement in safety and performance.

Medical Image Segmentation using U-Net (Personal Project)

- Implemented a U-Net architecture for semantic segmentation of medical images, achieving state-of-the-art results on a public dataset.
- Utilized data augmentation techniques to improve model robustness and generalization.
- Developed a web application for visualizing the segmentation results using React and JavaScript.

Facial Recognition System using FaceNet (Personal Project)

- Built a facial recognition system using FaceNet and TensorFlow, achieving high accuracy on a large dataset of facial images.
- Implemented a face verification system with a high level of security.
- Developed a user-friendly interface for managing and searching the facial database.